

Chapter 5: Thermochemistry

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CHM111
Lecture 5



Part I

Energy Basics

Outline of Part I: Energy Basics

What Is Energy?

The First Law of Thermodynamics

Work — Pressure-Volume (PV) Work

Outline of Part I: Energy Basics

What Is Energy?

The First Law of Thermodynamics

Work — Pressure-Volume (PV) Work

Energy

Definition: Energy

Energy is the capacity to do work or to produce heat.

Kinetic Energy

Energy of **motion**.

Examples:

- ▶ A moving car
- ▶ Flowing water
- ▶ Vibrating atoms in a warm object

Potential Energy

Energy of **position** or stored energy.

Examples:

- ▶ Water held behind a dam
- ▶ A compressed spring
- ▶ Chemical bonds in fuel

In chemistry, the most important form of potential energy is **chemical energy** — energy stored in chemical bonds.

Forms of Energy

Energy exists in many forms, all of which can be converted into one another:

- ▶ **Mechanical:**
energy of moving objects or stored in stretched/compressed materials
- ▶ **Radiant (electromagnetic):**
energy carried by light and other radiation
- ▶ **Nuclear:**
energy stored in atomic nuclei
- ▶ **Electrical:**
energy from moving electric charges
- ▶ **Chemical:**
energy stored in chemical bonds; released or absorbed during reactions
- ▶ **Thermal:**
total kinetic energy of all particles in a sample



The **transfer of energy** from one form to another appears as **work** and/or **heat**.

Energy Units

The SI unit of energy is the **joule** (J):

$$1 \text{ J} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$$

The Calorie

- **calorie (cal)**: amount of heat needed to raise 1 g of water by 1 °C.
- **Calorie (Cal)** = 1 kcal = food energy unit

$$1 \text{ cal} = 4.184 \text{ J}$$

$$1 \text{ Cal} = 1 \text{ kcal} = 4184 \text{ J}$$

A cheeseburger with 500 Cal contains

$$500 \text{ Cal} \times \frac{4184 \text{ J}}{1 \text{ Cal}} \approx 2.1 \times 10^6 \text{ J}$$

of chemical potential energy.

Check Your Understanding

Classify each as primarily **kinetic** or **potential** energy:

- (a) A ball rolling down a ramp
- (b) Gasoline sitting in a fuel tank
- (c) A stretched rubber band
- (d) Steam flowing through a pipe

Come to lecture to see the answer.

Outline of Part I: Energy Basics

What Is Energy?

The First Law of Thermodynamics

Work — Pressure-Volume (PV) Work

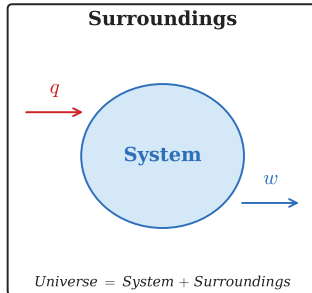
System and Surroundings

System

The part of the universe under study — usually the reacting chemicals.

Surroundings

Everything outside the system — the container, the room, etc.



Energy can be exchanged between system and surroundings in two ways:

- **Heat (q):** energy transferred due to a temperature difference
- **Work (w):** energy transferred by a force acting through a distance

The **universe** is defined as any system plus its surroundings.

Internal Energy and the First Law

Internal Energy (E or U)

The **internal energy** of a system is the sum of all its particles' kinetic and potential energies.

First Law of Thermodynamics (1LOT)

Energy is **neither created nor destroyed**; it is simply transferred from one form to another.

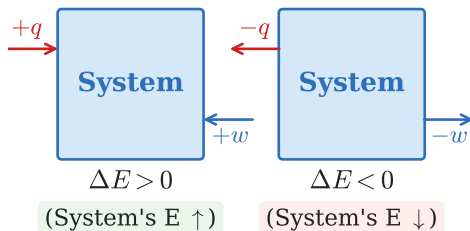
$$\Delta E_{\text{universe}} = \Delta E_{\text{system}} + \Delta E_{\text{surroundings}} = 0$$



The change in internal energy of a system equals the heat added to it plus the work done on it:

$$\Delta E = q + w$$

Sign Conventions for q and w



$$\Delta E = q + w$$

Heat (q)

- ▶ $q > 0$: heat flows **into** the system (endothermic)
- ▶ $q < 0$: heat flows **out of** the system (exothermic)

Work (w)

- ▶ $w > 0$: work done **on** the system (surroundings compress system)
- ▶ $w < 0$: work done **by** the system (system expands against surroundings)

Check Your Understanding

A chemical reaction releases heat to the surroundings and causes a gas to expand against a piston.

What are the signs of q and w for the **system**?

Is ΔE positive or negative? Is it large or small relative to q alone?

Come to lecture to see the answer.

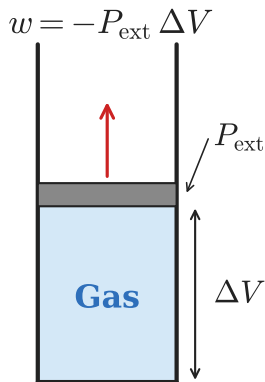
Outline of Part I: Energy Basics

What Is Energy?

The First Law of Thermodynamics

Work — Pressure-Volume (PV) Work

Pressure-Volume (PV) Work



Expansion: $\Delta V > 0 \Rightarrow w < 0$

PV Work

When a gas expands or contracts against a constant external pressure P , the work done *by* the system (the gas) is:

$$w = -P \Delta V$$

where $\Delta V = V_{\text{final}} - V_{\text{initial}}$.

Expansion ($\Delta V > 0$)

Gas pushes piston out.

$$w = -P \Delta V < 0$$

System *loses* energy by doing work on surroundings.

Contraction ($\Delta V < 0$)

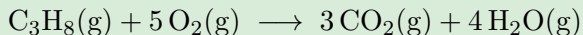
Piston compresses gas.

$$w = -P \Delta V > 0$$

System *gains* energy as surroundings do work on it.

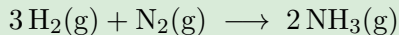
Changing the Amount of Gaseous Species Changes Volume

Expansion ($\Delta V > 0$): More moles of gas produced



6 mol gas $\rightarrow$ 7 mol gas $\therefore \Delta V > 0, w < 0$

Contraction ($\Delta V < 0$): Fewer moles of gas produced

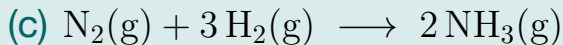
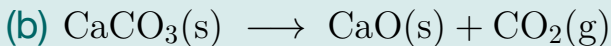
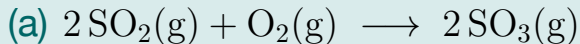


4 mol gas $\rightarrow$ 2 mol gas $\therefore \Delta V < 0, w > 0$

When counting moles of gas, **only** gas-phase species matter — solids and liquids do not contribute significantly to ΔV .

Check Your Understanding

For each reaction, predict the sign of w (work done on/by the system):



Come to lecture to see the answer.

Part II

Calorimetry

Outline of Part II: Calorimetry

Heat Capacity and Specific Heat

Calorimetry

Outline of Part II: Calorimetry

Heat Capacity and Specific Heat

Calorimetry

Heat Capacity

Heat Capacity (C)

The **heat capacity** of an object is the quantity of heat required to raise its temperature by 1°C (or 1 K).

$$q = C \Delta T$$

Heat capacity is an **extensive** property — it depends on the amount of material.

Specific Heat Capacity (c or c_s)

The **specific heat capacity** is the heat required to raise 1 g of a substance by 1°C . It is an **intensive** property.

$$q = m c \Delta T$$

Units: $\frac{\text{J}}{\text{g}^{\circ}\text{C}}$ or $\frac{\text{J}}{\text{g K}}$

Specific Heat Capacities of Common Substances

Substance	Specific Heat (c), $\frac{\text{J}}{\text{g}^\circ\text{C}}$
Water (liquid)	4.184
Water (ice)	2.093
Water (steam)	1.864
Aluminum	0.897
Iron	0.449
Copper	0.385
Gold	0.129

Water has an exceptionally **high** specific heat — it resists temperature change. This is why coastal climates are mild and why water is used in cooling systems.

Check Your Understanding

Two blocks of equal mass — one aluminum ($c = 0.897 \frac{\text{J}}{\text{g}^\circ\text{C}}$) and one gold ($c = 0.129 \frac{\text{J}}{\text{g}^\circ\text{C}}$) — are heated with the same amount of energy.

Which block reaches a higher temperature? Why?

Come to lecture to see the answer.

Example: Using $q = mc\Delta T$

Heating Iron

How much heat is necessary to raise the temperature of 1.00 g of iron from 17.0 °C to 27.0 °C?

$$(c_{\text{Fe}} = 0.449 \frac{\text{J}}{\text{g}^{\circ}\text{C}})$$

Given:

$$m = 1.00 \text{ g}$$

$$c = 0.449 \frac{\text{J}}{\text{g}^{\circ}\text{C}}$$

$$\Delta T = 27.0 - 17.0 = 10.0^{\circ}\text{C}$$

$$q = mc\Delta T$$

$$q = (1.00 \text{ g})(0.449 \frac{\text{J}}{\text{g}^{\circ}\text{C}})(10.0^{\circ}\text{C})$$

$$q = 4.49 \text{ J}$$

Check Your Understanding

A 55.0 g piece of copper ($c = 0.385 \frac{\text{J}}{\text{g}^\circ\text{C}}$) is heated from 25.0°C to 75.0°C .

How much heat does the copper absorb?

Come to lecture to see the answer.

Outline of Part II: Calorimetry

Heat Capacity and Specific Heat

Calorimetry

Calorimetry

Definition: Calorimetry

Calorimetry is the measurement of heat flow associated with a chemical or physical process.

Calorimeter

A **calorimeter** is an insulated device used to measure heat changes. Because the calorimeter is insulated, all heat from the reaction stays within the system:

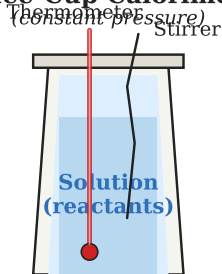
$$q_{\text{rxn}} = -q_{\text{calorimeter}}$$

The negative sign is because the heat *released by* the reaction equals the heat *absorbed by* the calorimeter (and vice versa).

Coffee-Cup Calorimetry

- ▶ Simple, open to atmosphere — measures at **constant pressure**.
- ▶ $q_{\text{rxn}} = -q_{\text{solution}} = -mc_s\Delta T$
- ▶ We assume $c_{\text{solution}} \approx c_{\text{water}} = 4.184 \frac{\text{J}}{\text{g}^\circ\text{C}}$
- ▶ Measures **enthalpy change** (ΔH) directly.

Coffee-Cup Calorimeter



$$q_{\text{rxn}} = -q_{\text{soln}} = -mc_s\Delta T$$

Sign check: if T goes up, reaction is exothermic ($q_{\text{rxn}} < 0$).

Example: Coffee-Cup Calorimetry

Dissolving NaOH

5.00 g of NaOH is dissolved in 100.0 g of water at 23.6 °C. The final temperature is 47.4 °C. Find q_{rxn} .

Given:

$$m = 100.0 + 5.00 = 105.0 \text{ g}$$

$$c = 4.184 \frac{\text{J}}{\text{g}^\circ\text{C}}$$

$$\Delta T = 47.4 - 23.6 = 23.8^\circ\text{C}$$

$$\begin{aligned} q_{\text{soln}} &= mc\Delta T \\ &= (105.0)(4.184)(23.8) \\ &= 10\,450 \text{ J} \end{aligned}$$

$$q_{\text{rxn}} = -10\,450 \text{ J}$$

$$q_{\text{rxn}} \approx -10.5 \text{ kJ}$$

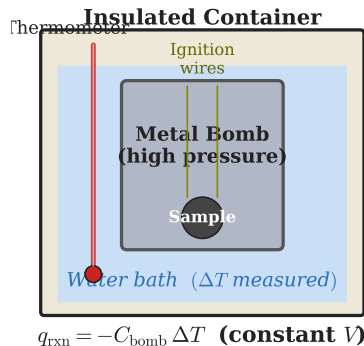
$q_{\text{rxn}} < 0$ confirms this is an **exothermic** process (dissolving NaOH releases heat).

Bomb Calorimetry

- ▶ Sealed, rigid vessel — measures at **constant volume**.
- ▶ Specifically used for combustion reactions.
- ▶ The bomb has a known heat capacity C_{bomb} :

$$q_{\text{rxn}} = -C_{\text{bomb}} \Delta T$$

- ▶ Measures ΔE (internal energy), not ΔH .
- ▶ $\Delta H \approx \Delta E$ for most combustion reactions.



Coffee-cup: ΔH ; bomb: ΔE .
 $\Delta H \approx \Delta E$ for most combustion reactions.

Check Your Understanding

When 1.00 g of benzoic acid is burned in a bomb calorimeter, the temperature rises by 2.57°C . The calorimeter has a heat capacity of $3.34 \frac{\text{kJ}}{^{\circ}\text{C}}$.

What is q_{rxn} (the heat of the reaction)?

Come to lecture to see the answer.

Part III

Enthalpy

Outline of Part III: Enthalpy

Enthalpy and ΔH

Thermochemical Equations

Enthalpy of Combustion

Outline of Part III: Enthalpy

Enthalpy and ΔH

Thermochemical Equations

Enthalpy of Combustion

Enthalpy (H)

Definition: Enthalpy

Enthalpy (H) is a thermodynamic state function defined as:

$$H = E + PV$$

where E is internal energy, P is pressure, and V is volume.

At **constant pressure**:

$$\begin{aligned}\Delta H &= \Delta E + P\Delta V \\ &= \Delta E - w \quad (\because w = -P\Delta V) \\ &= (q + w) - w\end{aligned}$$

$$\therefore \Delta H = q_P$$

The enthalpy change equals the **heat exchanged at constant pressure**.

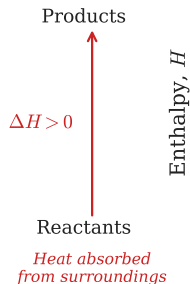
Endothermic and Exothermic Reactions

Endothermic ($\Delta H > 0$)

System **absorbs** heat.

Products have *more* enthalpy.

E.g., melting ice, photosynthesis

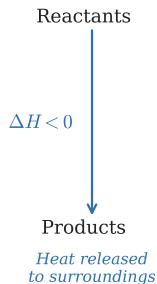


Exothermic ($\Delta H < 0$)

System **releases** heat.

Products have *less* enthalpy.

E.g., combustion, dissolving NaOH



$\Delta H > 0 \Rightarrow$ endothermic

$\Delta H < 0 \Rightarrow$ exothermic

Check Your Understanding

Classify each process as **endothermic** or **exothermic**:

- (a) Burning wood in a fireplace
- (b) An ice pack becoming cold when activated
- (c) Condensation of water vapor on a cold glass
- (d) Baking a cake

Come to lecture to see the answer.

Outline of Part III: Enthalpy

Enthalpy and ΔH

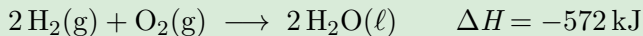
Thermochemical Equations

Enthalpy of Combustion

Thermochemical Equations

Definition

A **thermochemical equation** is a balanced chemical equation that also states the enthalpy change ΔH of the reaction.

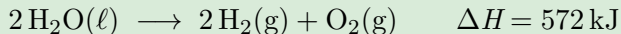


Notes on Thermochemical Equations:

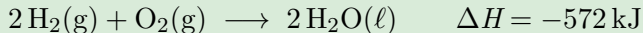
- ▶ ΔH refers to the equation **as written** (with the given stoichiometric coefficients).
- ▶ State symbols (g, l, s, aq) are **required** — enthalpy depends on phase.
- ▶ Reversing the reaction **changes the sign** of ΔH .
- ▶ Multiplying the equation by a factor **multiplies** ΔH by the same factor.

Thermochemical Equations: Reverse and Scale

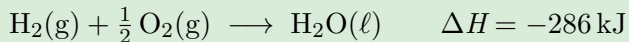
Original



Reversed (sign of ΔH changes)

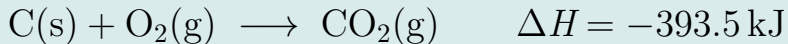


Scaled by $\frac{1}{2}$ (ΔH also scaled by $\frac{1}{2}$)

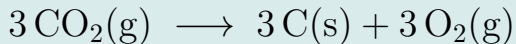


Check Your Understanding

Given:



What is ΔH for the following reaction?



Come to lecture to see the answer.

Outline of Part III: Enthalpy

Enthalpy and ΔH

Thermochemical Equations

Enthalpy of Combustion

Standard Enthalpy of Combustion

Definition

The **standard enthalpy of combustion** (ΔH_c°) is the enthalpy change when 1 mol of a substance burns completely in oxygen under standard conditions (25 °C, 1 atm).

Substance	$\Delta H_c^\circ \left(\frac{\text{kJ}}{\text{mol}} \right)$
Hydrogen, $\text{H}_2(\text{g})$	−286
Carbon (graphite), $\text{C}(\text{s})$	−394
Methane, $\text{CH}_4(\text{g})$	−890
Propane, $\text{C}_3\text{H}_8(\text{g})$	−2220
Ethanol, $\text{C}_2\text{H}_5\text{OH}(\text{l})$	−1367

Combustion enthalpies are always **negative** (exothermic).

Part IV

Hess's Law & Standard Enthalpies

Outline of Part IV: Hess's Law & Standard Enthalpies

Hess's Law

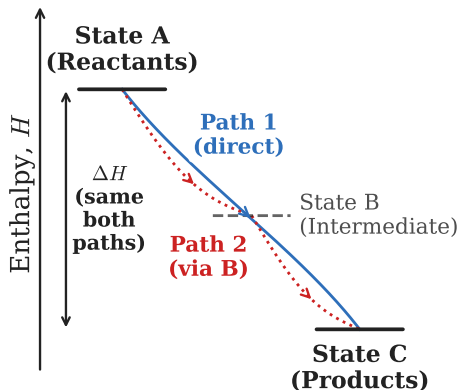
Standard Enthalpy of Formation

Outline of Part IV: Hess's Law & Standard Enthalpies

Hess's Law

Standard Enthalpy of Formation

State Functions and Path Independence



ΔH depends only on initial and final states

Enthalpy is a State Function

The value of a **state function** depends only on the **initial and final states** of a system, *not* on the path taken between them.

Energy, temperature, and enthalpy are state functions.

Heat (q) and work (w) are *not*.

Analogy: The elevation change when hiking from a valley to a summit is the same regardless of which trail you take. Elevation is a state function; the distance walked is not.

Hess's Law

Hess's Law

If a reaction can be expressed as the sum of two or more reactions, then ΔH for the overall reaction equals the **sum** of the ΔH values of those reactions.

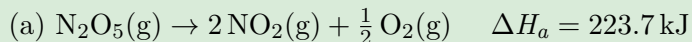
Procedure

1. Identify reactions whose sum gives the target reaction.
2. **Reverse** any reaction as needed (change sign of ΔH).
3. **Scale** any reaction as needed (multiply ΔH by same factor).
4. Add the reactions and their ΔH values; cancel species that appear on both sides.

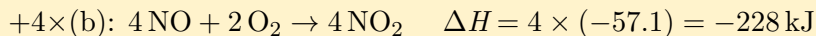
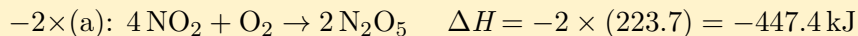
Example: Hess's Law

Finding ΔH for $4 \text{ NO(g)} + 3 \text{ O}_2\text{(g)} \rightarrow 2 \text{ N}_2\text{O}_5\text{(g)}$

Given:



 Strategy: reverse (a) twice; add (b) four times.

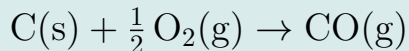


$$\begin{aligned}\Delta H_{\text{rxn}} &= -2(\Delta H_a) + 4(\Delta H_b) \\ &= -2(223.7) + 4(-57.1)\end{aligned}$$

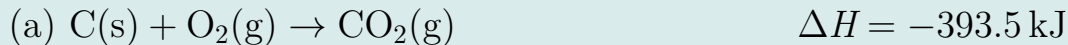
$$\Delta H_{\text{rxn}} = -447.4 - 228 = -676 \text{ kJ}$$

Check Your Understanding

Use Hess's Law to find ΔH for:



Given:



Come to lecture to see the answer.

Outline of Part IV: Hess's Law & Standard Enthalpies

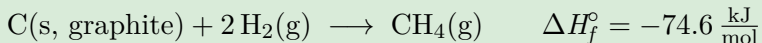
Hess's Law

Standard Enthalpy of Formation

Standard Enthalpy of Formation (ΔH_f°)

Definition

The **standard enthalpy of formation** (ΔH_f°) is the enthalpy change when 1 mol of a compound is formed from its **constituent elements in their standard states** at 25 °C and 1 atm.



By definition, the standard enthalpy of formation of any **element in its standard state** is exactly **zero**:

$$\Delta H_f^\circ[\text{O}_2(\text{g})] = 0, \quad \Delta H_f^\circ[\text{C(s, graphite)}] = 0, \quad \Delta H_f^\circ[\text{Fe(s)}] = 0$$

Standard Enthalpies of Formation: Selected Values

Compound	$\Delta H_f^\circ \left(\frac{\text{kJ}}{\text{mol}} \right)$
H ₂ O(l)	−286.8
H ₂ O(g)	−241.8
CO ₂ (g)	−393.5
CH ₄ (g) (methane)	−74.6
NH ₃ (g)	−46.1
NO ₂ (g)	+33.2
C ₆ H ₆ (l) (benzene)	+49.1

Most compounds have *negative* ΔH_f° — they are thermodynamically more stable than their elements. A *positive* ΔH_f° (e.g. NO₂, benzene) means the compound is less stable than its elements.

Calculating $\Delta H_{\text{rxn}}^{\circ}$ from Formation Enthalpies

Hess's Law Applied to Formation Enthalpies

$$\Delta H_{\text{rxn}}^{\circ} = \sum n \Delta H_f^{\circ}(\text{products}) - \sum m \Delta H_f^{\circ}(\text{reactants})$$

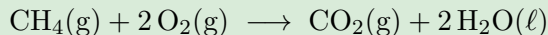
where n and m are stoichiometric coefficients.

Strategy: “products minus reactants,” each multiplied by its stoichiometric coefficient.

Example: $\Delta H_{\text{rxn}}^{\circ}$ from ΔH_f°

Combustion of Methane

Calculate $\Delta H_{\text{rxn}}^{\circ}$ for:



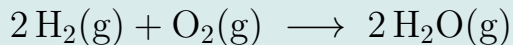
Use: $\Delta H_f^{\circ}[\text{CH}_4(\text{g})] = -74.6 \frac{\text{kJ}}{\text{mol}}$, $\Delta H_f^{\circ}[\text{CO}_2(\text{g})] = -393.5 \frac{\text{kJ}}{\text{mol}}$,
 $\Delta H_f^{\circ}[\text{H}_2\text{O}(\ell)] = -286.8 \frac{\text{kJ}}{\text{mol}}$, $\Delta H_f^{\circ}[\text{O}_2(\text{g})] = 0$

$$\begin{aligned}\Delta H_{\text{rxn}}^{\circ} &= [1(-393.5) + 2(-286.8)] - [1(-74.6) + 2(0)] \\ &= [-393.5 - 573.6] - [-74.6]\end{aligned}$$

$$\Delta H_{\text{rxn}}^{\circ} = -892.5 \text{ kJ}$$

Check Your Understanding

Calculate $\Delta H_{\text{rxn}}^{\circ}$ for the formation of water vapor from its elements:



Use $\Delta H_f^{\circ}[\text{H}_2\text{O}(\text{g})] = -241.8 \frac{\text{kJ}}{\text{mol}}$.

Come to lecture to see the answer.

Check Your Understanding

Rank these processes from **most endothermic** to **most exothermic** (no calculation needed):

- (a) Burning hydrogen: $\Delta H = -286 \frac{\text{kJ}}{\text{mol}}$
- (b) Dissolving ammonium nitrate: $\Delta H = 25 \frac{\text{kJ}}{\text{mol}}$
- (c) Combustion of propane: $\Delta H = -2220 \frac{\text{kJ}}{\text{mol}}$
- (d) Formation of NO_2 : $\Delta H = 33 \frac{\text{kJ}}{\text{mol}}$

Come to lecture to see the answer.

Summary: Chapter 5

- ▶ **Energy** exists in many forms; transfers appear as heat (q) or work (w).
- ▶ **First Law:** $\Delta E = q + w$; energy is conserved; $\Delta E_{\text{universe}} = 0$.
- ▶ **PV work:** $w = -P\Delta V$; expansion does work on surroundings ($w < 0$).
- ▶ **Specific heat:** $q = mc\Delta T$; water has an unusually high c ($4.184 \frac{\text{J}}{\text{g}^\circ\text{C}}$).
- ▶ **Calorimetry:** $q_{\text{rxn}} = -q_{\text{calorimeter}}$; coffee-cup measures ΔH ; bomb measures ΔE .
- ▶ **Enthalpy:** $H = E + PV$; at constant pressure, $\Delta H = q_P$.
- ▶ **Endothermic** ($\Delta H > 0$) vs. **Exothermic** ($\Delta H < 0$).
- ▶ **Hess's Law:** ΔH is a state function; ΔH_{rxn} is path-independent.
- ▶ **Standard enthalpy of formation** (ΔH_f°): enthalpy to form 1 mol of compound from elements in standard states; elements have $\Delta H_f^\circ = 0$.
- ▶ $\Delta H_{\text{rxn}}^\circ = \sum n \Delta H_f^\circ(\text{products}) - \sum m \Delta H_f^\circ(\text{reactants})$.